DATA ANALYTICS USING IT PROTOCOL DATASET

A SUMMER INTERNSHIP REPORT

Submitted by

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BONAFIDE CERTIFICATE

This is to certify this Summer Internship project report "**DATA ANALYTICS USING IT PROTOCOL DATASET**" is the bonafide work of **P.KEERTHANA(810023205065)** has satisfactory completed the course of **CS3711 - SUMMER INTERSHIP** for **SEVENTH** semester, **B.TECH INFORMATION TECHNOLOGY** during the academic year 2026-2027.

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Submitted for the practical examination held on _____

ABSTRACT

This internship was carried out in the field of **Data Analytics**, with the objective of gaining practical knowledge in data management, analysis, and visualization using industry-standard tools. The internship involved working with an **IT Protocol Dataset** to understand the complete data analytics workflow, from database management to dashboard development.

After organizing the data, **Microsoft Excel** was used for data cleaning and preprocessing. Duplicate records were removed, missing values were handled, and the dataset was prepared for analysis. Pivot Tables, Pivot Charts, slicers, and KPI cards were used to create an interactive dashboard that displayed important metrics such as firewall activity, connection loss, CPU usage, memory usage, threat scores, and department-wise performance.

The cleaned dataset was then imported into **Microsoft Power BI**, where Power Query was used for data transformation and DAX was used to create calculated measures. Interactive dashboards were developed using various visualizations, including bar charts, line charts, pie charts, donut charts, funnel charts, KPI cards, and slicers. These dashboards provided valuable insights into bandwidth utilization, software distribution, compliance status, network environments, license types, monthly costs, and protocol analysis.

The project began with **MySQL Workbench**, where SQL was used to create databases and tables, insert, update, delete, and retrieve data using various SQL queries. Concepts such as joins, aggregate functions, grouping, subqueries, and relationships were implemented to efficiently manage and analyze the dataset.

The internship enhanced my knowledge of SQL, data cleaning, data visualization, dashboard design, and business intelligence. It also improved my analytical thinking, problem-solving abilities, and technical skills in handling real-world datasets. Overall, this internship provided valuable practical experience in Data Analytics and prepared me for future opportunities in the fields of Data Analysis and Business Intelligence.

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CHAPTER 1
INTRODUCTION & OBJECTIVES

1. INTRODUCTION

Data Analytics is the process of collecting, organizing, cleaning, transforming, analyzing, and visualizing data to discover useful information that supports effective decision-making. In today's digital world, organizations generate large volumes of data from various business operations.

As part of my internship, I worked on an **IT Protocol Dataset** that contains information related to network infrastructure and IT operations. The dataset includes details such as protocol types, network environments, software publishers, software license types, bandwidth usage, CPU cores, monthly costs, compliance status, departments, operating systems, and device types.

The project was completed in three major phases.

The first phase involved **database management using MySQL Workbench**. A relational database was created, and SQL was used to perform operations such as creating tables, inserting records, updating data, deleting unwanted records, modifying table structures, and retrieving information using SQL queries. Advanced SQL concepts including joins, aggregate functions, GROUP BY, HAVING clauses, and subqueries were also applied to analyze the data efficiently.

The second phase focused on **Microsoft Excel**. The dataset was cleaned by removing duplicate records, correcting formatting issues, handling missing values, and organizing the data into a structured format. Pivot Tables, Pivot Charts, slicers, and KPI cards were used to develop an interactive dashboard. The Excel dashboard presents key metrics such as firewall activity, connection loss, CPU usage, memory usage, threat scores, packet loss percentage, and department-wise analysis.

The third phase involved **Microsoft Power BI**, where the cleaned dataset was imported for advanced visualization. Power Query was used for data transformation, and DAX (Data Analysis Expressions) was used to create calculated measures. Various interactive visualizations such as bar charts, line charts, donut charts, pie charts, funnel charts, KPI cards, and slicers were used to create a professional dashboard. The Power BI dashboard provides insights into bandwidth utilization, software distribution, compliance status, network environments, software licensing, and monthly operational costs.

The dashboards developed during this internship help users analyze IT infrastructure efficiently through interactive visualizations and filtering options. These reports enable organizations to monitor system performance, identify trends, compare departmental performance, and support data-driven decision-making.

Overall, this internship provided valuable practical experience in **SQL database management, data cleaning, dashboard development, business intelligence, and data visualization**. It enhanced my technical knowledge and strengthened my analytical and problem-solving abilities, preparing me for future opportunities in the field of Data Analytics and Business Intelligence.

1.1 OBJECTIVES OF THE INTERNSHIP

The primary objective of this internship was to gain practical knowledge of **Data Analytics** by working with real-world datasets and modern analytical tools. The internship aimed to bridge the gap between academic concepts and industry practices.

The specific objectives of the internship are as follows:

- To understand the fundamentals of Data Analytics and Business Intelligence.
- To learn database management using MySQL Workbench.
- To create databases, tables, and relationships using SQL.
- To perform SQL operations such as CREATE, INSERT, UPDATE, DELETE, ALTER, and SELECT.

- To retrieve meaningful information using SQL queries, joins, aggregate functions, GROUP BY, HAVING, and subqueries.
- To clean and preprocess the dataset before analysis.
- To create interactive dashboards using Microsoft Excel.
- To use Pivot Tables, Pivot Charts, slicers, and KPI cards for data visualization.
- To develop professional dashboards using Microsoft Power BI.
- To perform data transformation using Power Query.
- To create calculated columns and measures using DAX.
- To analyze the IT Protocol Dataset and identify meaningful business insights.
- To improve analytical thinking, problem-solving, and decision-making skills.

1.2.COMPAMY PROFILE



ADDRESS

VDart Academy

No. 30, Chennai–Theni Highway,

Mannarpuram,

Trichy – 620020.

Contact: 8870397119

Website: www.vdart.com

Company Profile

VDart, founded in **2007**, is a global **IT staffing and digital transformation company** offering talent management, technology solutions, and consulting services. Since its inception, the company has expanded its operations across several countries, including the **United States, Canada, the United Kingdom, and India**, serving industries such as automotive, banking, healthcare, and manufacturing.

The company provides a wide range of services, including application development and maintenance, ERP implementation and consulting, software testing, performance engineering, software localization and globalization, IT infrastructure management, Business Process Outsourcing (BPO), and IT education and training.

With its business philosophy of **"Beyond Technology"**, VDart is committed to delivering innovative information technology solutions that support the sustainable growth of businesses and society. The organization strives to be recognized as a trusted company by employees, customers, shareholders, and the communities it serves.

The office is located at **40, Raja Colony, Collector Office Road, Tiruchirappalli – 620001**, with additional operations at the **Global Capability Center (GCC)** in Mannarpuram on the Chennai–Theni Highway, Trichy – 620020.

MISSION

The mission of **VDart Academy** includes:

1. Providing high-quality software development services, professional consulting, and development outsourcing to improve customers' business operations.
2. Making access to information easier and more secure for enterprise businesses.
3. Improving communication and data exchange through Business-to-Business (B2B) solutions.

4. Delivering value-for-money services while maintaining high standards of quality and customer satisfaction.

1.3.DAILY WORK REPORT

DAY 1:

On the first day, as the fantasy solution, I went to the concerned and and conversed with the concern forehead about my internship. I chose to pursue an academic career in Full Stack Development.

DAY 2:

On the this day, I have learnt topic related to Data Analytics, The topics are, introduction to Data Analytics, Phases of Data Analytics project, Data Cleaning, Functions ,Sort and filter, Lookup functions, Conditional formatting, Data Validation .

DAY 3:

On the third day , I have finish the great learning lesson and activity.I have learnt how to create the pivot table.I have learnt basic functions in dataset.

DAY 4:

On the forth day, I have learnt the advanced functions and data cleaning in depth concept.In all I learnt syntax and working with large dataset.

DAY 5:

On the fifth day, I have create the pivot table and I have create the Dashboard for my dataset.

DAY 6:

On the sixth day, I have learnt about Kpi cards and Macros.I will attach in my dashboard.

DAY 7:

On the seventh day,I have learnt about drop-down and worked with example to create another dashboard for drop-down.

DAY 8:

On the eighth day, I learnt about power bi basic, and I finish my dashboard in excel sheet

DAY 9:

On the Nineth day, I have start the dashboard in Power BI.

DAY 10:

On the Tenth day, I have learnt about Power BI in Great learning (Data visualization using Power BI)

DAY 11:

On the Eleventh day, I complete my dashboard in excel sheet . And learn about power Bi.

DAY 12:

On the Twelfth day, I will modified the mistake in my dashboard. Recheck the insert and pivot table in my dashboard.

DAY 13:

On the Thirteenth day, I have finish my dashboard and dropdown list . Next move to the Power BI.

DAY 14:

On the fourteenth day,I learnt about the basic of the Power BI.Then I get my dataset and transform it to cleaning the dataset and load the dataset in Power BI.

DAY 15:

On the fifteenth day, I have finished the Dashboard such as charts,title cards in Power BI.

DAY 16:

On the sixteenth day,I have finished the KPI cards and slicer for my Power BI dashboard.

DAY 17:

On the seventeenth day, I have finished Power BI for my Dataset.

DAY 18:

On the Eighteenth day, I have learnt about numpy and I finish my dashboard in Power BI for my dataset.

DAY 19:

On the Nineteenth day,I have learn about SQL concepts as basic functions.

DAY 20:

On the Twentieth day, I have learnt and implemented the SQL queries in the workbench.

DAY 21:

On the Twenty-first day, I have worked in MySQL like joints and subqueries.

CHAPTER 2
LITERATURE REVIEW / SYSTEM STUDY

2.1 Evaluation of Existing System

In today's digital environment, organizations generate a large volume of data from their daily operations. This data contains valuable information that can support decision-making and improve organizational performance. However, many organizations still rely on traditional reporting methods, such as spreadsheets and manually prepared reports, to analyze and present data. These methods become inefficient when dealing with large and complex datasets.

Traditional reporting systems require manual data entry, frequent updates, and repetitive calculations. As the amount of data increases, maintaining accuracy and consistency becomes difficult. Manual reporting also consumes significant time and increases the possibility of human errors. In addition, traditional reports offer limited visualization capabilities, making it difficult for users to identify trends, patterns, and business insights.

Database Management Systems (DBMS) such as **MySQL** provide an efficient way to store, organize, and retrieve structured data using SQL queries. SQL enables users to perform data manipulation, filtering, aggregation, and analysis more efficiently than manual methods.

Similarly, modern Business Intelligence (BI) tools such as **Microsoft Power BI** and **Microsoft Excel** provide advanced visualization features that help organizations convert raw data into meaningful information. These tools support interactive dashboards, charts, graphs, filters, and Key Performance Indicators (KPIs), enabling users to analyze business performance quickly and effectively.

Compared with traditional reporting methods, Business Intelligence tools provide improved accuracy, faster report generation, better visualization, and enhanced decision-making capabilities.

2.2 Problem Statement Identification

The IT Protocol Dataset contains information related to various aspects of an organization's IT infrastructure, including network protocols, software applications, departments, bandwidth usage, CPU utilization, software licensing, operating systems, firewall activities, and monthly operational costs. Managing and analyzing such a large dataset manually is challenging and time-consuming.

Several problems were identified in the existing system:

- Data is stored in multiple formats, making analysis difficult.
- Manual report preparation requires considerable time and effort.
- Large datasets are difficult to analyze using conventional methods.
- Identifying relationships between different data fields is challenging.
- Traditional reports do not provide interactive filtering and visualization.
- Decision-makers cannot easily identify trends, patterns, or anomalies from static reports.
- Manual calculations increase the possibility of data inconsistency and human errors.
- Monitoring IT resources and network performance becomes difficult without proper dashboards.

2.3 Proposed Solution / Methodology

To overcome the limitations of the existing system, a Data Analytics solution was developed using **MySQL Workbench, Microsoft Excel, and Microsoft Power BI**.

The project follows a systematic methodology consisting of the following phases:

Data Collection

The IT Protocol Dataset was collected, containing information related to network protocols, devices, departments, operating systems, software details, CPU cores,

bandwidth utilization, compliance status, software licensing, and monthly operational costs.

Database Management Using SQL

The dataset was organized using MySQL Workbench. SQL was used to create databases and tables, insert records, update data, delete unnecessary records, modify table structures, and retrieve required information using SQL queries. Various SQL concepts such as joins, aggregate functions, GROUP BY, HAVING, ORDER BY, and subqueries were applied to analyze the data effectively.

Data Cleaning and Preprocessing

Before visualization, the dataset was cleaned using Microsoft Excel. Duplicate records were removed, missing values were handled, incorrect data formats were corrected, and the dataset was standardized to improve data quality and consistency.

Dashboard Development Using Microsoft Excel

After cleaning the data, Pivot Tables and Pivot Charts were created in Microsoft Excel. Slicers, filters, and KPI cards were added to develop an interactive dashboard. The Excel dashboard displays important information such as firewall activity, connection loss, threat scores, CPU usage, memory usage, packet loss percentage, department-wise performance, operating systems, and protocol analysis.

Dashboard Development Using Microsoft Power BI

The cleaned dataset was imported into Microsoft Power BI for advanced visualization. Power Query was used for data transformation, and DAX (Data Analysis Expressions) was used to create calculated measures. Interactive dashboards were developed using bar charts, pie charts, donut charts, line charts, funnel charts, KPI cards, tables, and slicers. The dashboard provides insights into bandwidth utilization, software distribution, compliance status, network environments, software licensing, and monthly operational costs.

Result Analysis

The dashboards developed using Excel and Power BI enable users to analyze the IT Protocol Dataset efficiently. Interactive filters and visualizations help identify trends, compare departmental performance, monitor network usage, and support data-driven decision-making. The proposed solution significantly reduces manual effort, improves reporting accuracy, and enhances the overall efficiency of business analysis.

CHAPTER 3
SYSTEM REQUIREMENTS & ARCHITECTURE

3.1 Hardware and Software Requirements

To successfully complete the Data Analytics project, both hardware and software resources were required. The hardware provided sufficient processing power to handle data processing and dashboard creation, while the software tools were used for database management, data cleaning, analysis, visualization, and report generation.

3.1.1 Hardware Requirements

Component	Specification
Processor	Intel Core i5 (10th Generation) or above
RAM	8 GB or higher
Storage	256 GB SSD or higher
Operating System	Windows 10 / Windows 11
Monitor	14-inch or above
Internet Connection	Required for software updates and online resources

The above hardware configuration provided a smooth working environment for SQL query execution, Excel data analysis, and Power BI dashboard development.

3.1.2 Software Requirements

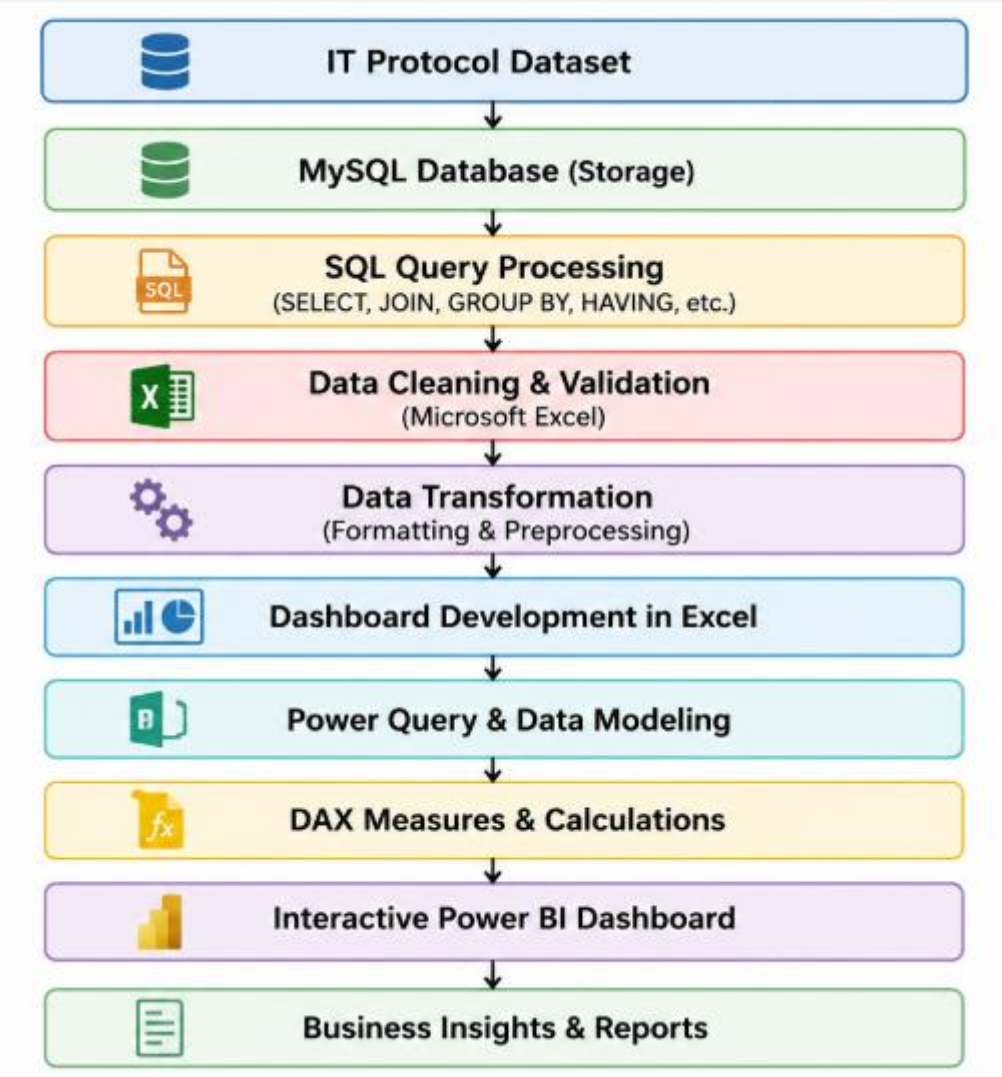
Software	Purpose
MySQL Workbench	Database creation and SQL query execution
MySQL Server	Database Management System
Microsoft Excel (Microsoft 365 / Excel 2019)	Data cleaning, Pivot Tables, Pivot Charts, Dashboard creation
Microsoft Power BI Desktop	Data visualization and dashboard development
Windows 10 / Windows 11	Operating System

These software tools were used throughout the internship to perform different stages of the data analytics process.

3.2 System Architecture

The system architecture illustrates the complete workflow followed during the internship project. The project consists of several stages, beginning with data collection and ending with dashboard generation and business insights.

System Architecture:



Architecture Description

The IT Protocol Dataset is initially stored in a MySQL database. SQL queries are used to retrieve, filter, and organize the required data. The retrieved data is then imported into Microsoft Excel, where data cleaning and preprocessing activities such as removing duplicate records, correcting data formats, and handling missing values are performed.

After preparing the dataset, interactive dashboards are created in Microsoft Excel using Pivot Tables, Pivot Charts, slicers, and KPI cards. The cleaned dataset is then imported into Microsoft Power BI for advanced visualization. Power Query is used for data transformation, while DAX is used to create calculated measures and key performance indicators. Finally, interactive dashboards provide meaningful business insights and support effective decision-making.

3.3 Workflow of the Data Analytics Project

The workflow followed during the internship is illustrated below.

Step 1: Data Collection

The IT Protocol Dataset containing information related to network devices, software, operating systems, protocols, departments, bandwidth usage, CPU utilization, software licensing, and monthly costs was collected.

Step 2: Database Management

The dataset was organized using MySQL Workbench. SQL commands were used to create databases and tables, insert records, modify data, and retrieve information through queries.

Step 3: Data Cleaning

The dataset was imported into Microsoft Excel. Duplicate values, missing records, and inconsistent data formats were identified and corrected to improve data quality.

Step 4: Data Analysis

The cleaned data was analyzed using SQL queries, Pivot Tables, and Pivot Charts to summarize and understand key metrics.

Step 5: Excel Dashboard Development

An interactive Excel dashboard was created using charts, slicers, KPI cards, and Pivot Tables. The dashboard displayed information such as firewall activity, connection loss, CPU usage, memory usage, threat scores, and packet loss percentage.

Step 6: Power BI Dashboard Development

The cleaned dataset was imported into Power BI Desktop. Power Query was used to transform the data, while DAX measures were created to calculate important business metrics. Interactive visualizations such as bar charts, line charts, pie charts, donut charts, funnel charts, and KPI cards were developed.

Step 7: Result Generation

The dashboards provided valuable insights into protocol distribution, software licensing, bandwidth utilization, compliance status, operating systems, network environments, and monthly operational costs. These insights help users monitor IT infrastructure and support data-driven decision-making

CHAPTER 4
DEVELOPMENT AND RESULT ANALYSIS

4.Development

The development phase of the internship involved collecting, organizing, analyzing, and visualizing the IT Protocol Dataset using MySQL Workbench, Microsoft Excel, and Microsoft Power BI. The project followed a systematic workflow to transform raw data into meaningful business insights.

4.1 Database Development using MySQL Workbench

The first stage of the project involved creating and managing the database using MySQL Workbench. A relational database was created to store the IT Protocol Dataset in an organized manner. SQL commands were used to create tables, define primary keys, and establish relationships between tables.

Various SQL operations were performed throughout the project, including:

- Creating databases and tables using CREATE statements.
- Inserting records using INSERT statements.
- Updating existing records using UPDATE statements.
- Deleting unnecessary records using DELETE statements.
- Retrieving data using SELECT queries.
- Sorting records using ORDER BY.
- Filtering records using WHERE clauses.
- Grouping data using GROUP BY and HAVING clauses.
- Applying aggregate functions such as COUNT(), SUM(), AVG(), MAX(), and MIN().
- Performing INNER JOIN, LEFT JOIN, and RIGHT JOIN operations.
- Using subqueries for advanced data retrieval.

These SQL operations ensured that the data was accurate, structured, and ready for further analysis.

4.1.2 Data Cleaning using Microsoft Excel

After extracting the required data from the database, the dataset was imported into Microsoft Excel for cleaning and preprocessing.

The following data preparation activities were performed:

- Removal of duplicate records.
- Identification and handling of missing values.
- Correction of inconsistent data formats.
- Standardization of column names.
- Verification of data accuracy.
- Data sorting and filtering.

These preprocessing steps improved the quality of the dataset and ensured reliable analysis.

4.2 Excel Dashboard Development

Once the dataset was cleaned, an interactive dashboard was developed in Microsoft Excel.

The dashboard was created using:

- Pivot Tables
- Pivot Charts
- Slicers
- Timeline Filters
- Conditional Formatting
- KPI Cards

The Excel dashboard displays important performance indicators such as:

- Firewall Action Distribution
- Connection Loss
- CPU Usage
- Memory Usage
- Threat Score
- Packet Loss Percentage
- Department-wise Analysis
- Operating System Distribution
- Protocol Distribution

The use of slicers allows users to interactively filter data based on different categories, making the dashboard user-friendly and informative.

4.3 Power BI Dashboard Development

The cleaned dataset was then imported into Microsoft Power BI Desktop for advanced data visualization.

Power Query Editor was used to transform the data by removing unnecessary columns, changing data types, and preparing the dataset for analysis. Relationships between tables were established where required, and DAX (Data Analysis Expressions) measures were created to calculate key performance indicators.

The Power BI dashboard includes:

- KPI Cards
- Clustered Bar Charts
- Line Charts
- Pie Charts
- Donut Charts
- Funnel Charts

- Tables
- Slicers
- Interactive Filters

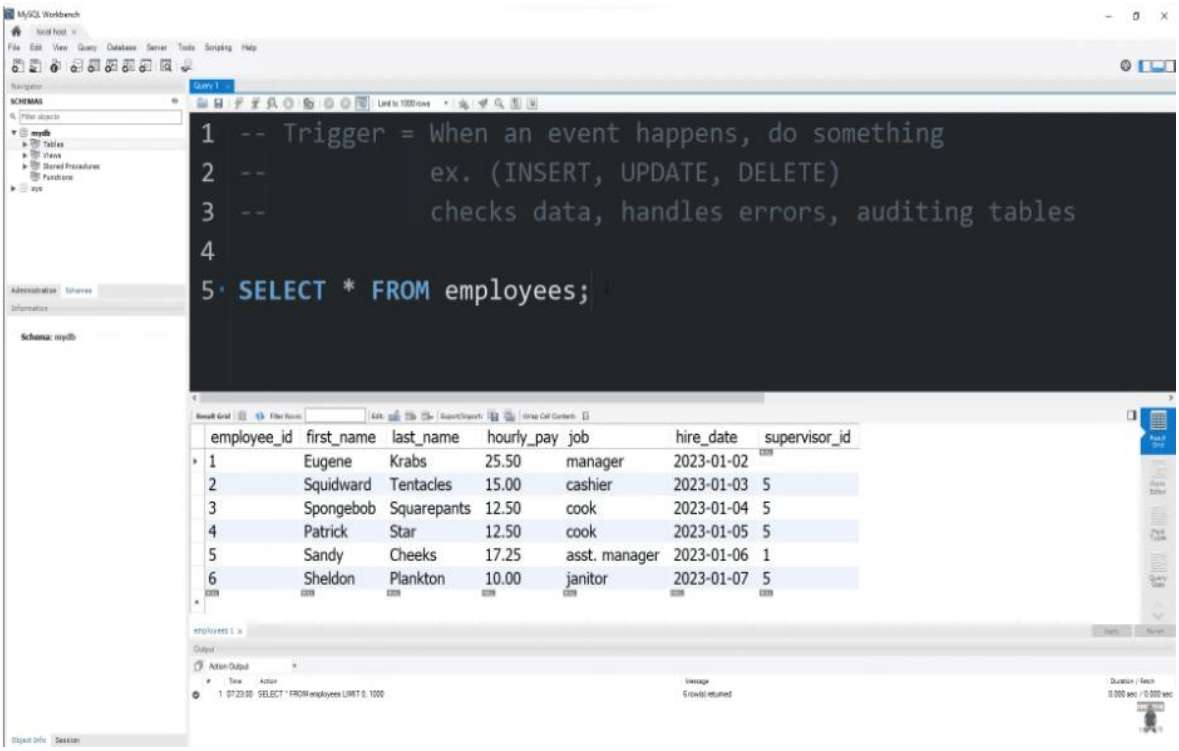
The dashboard presents important metrics such as Total CPU Cores, Software Cost, Current Asset Value, Monthly Cost, License Types, Compliance Status, Network Environment, Publisher-wise Software Distribution, and Bandwidth Utilization.

Power BI enables users to interact with the dashboard, filter data dynamically, and gain meaningful insights into IT infrastructure performance.

4.4 Result Analysis

The developed dashboards successfully transformed the raw IT Protocol Dataset into meaningful visual reports that support effective decision-making. The integration of SQL, Microsoft Excel, and Power BI provided a complete data analytics workflow from data storage to business intelligence reporting.

SQL Analysis



SQL queries helped retrieve accurate information from the database and simplified the analysis process. Aggregate functions, joins, and filtering techniques enabled efficient extraction of useful information from large datasets.

Excel Dashboard Analysis

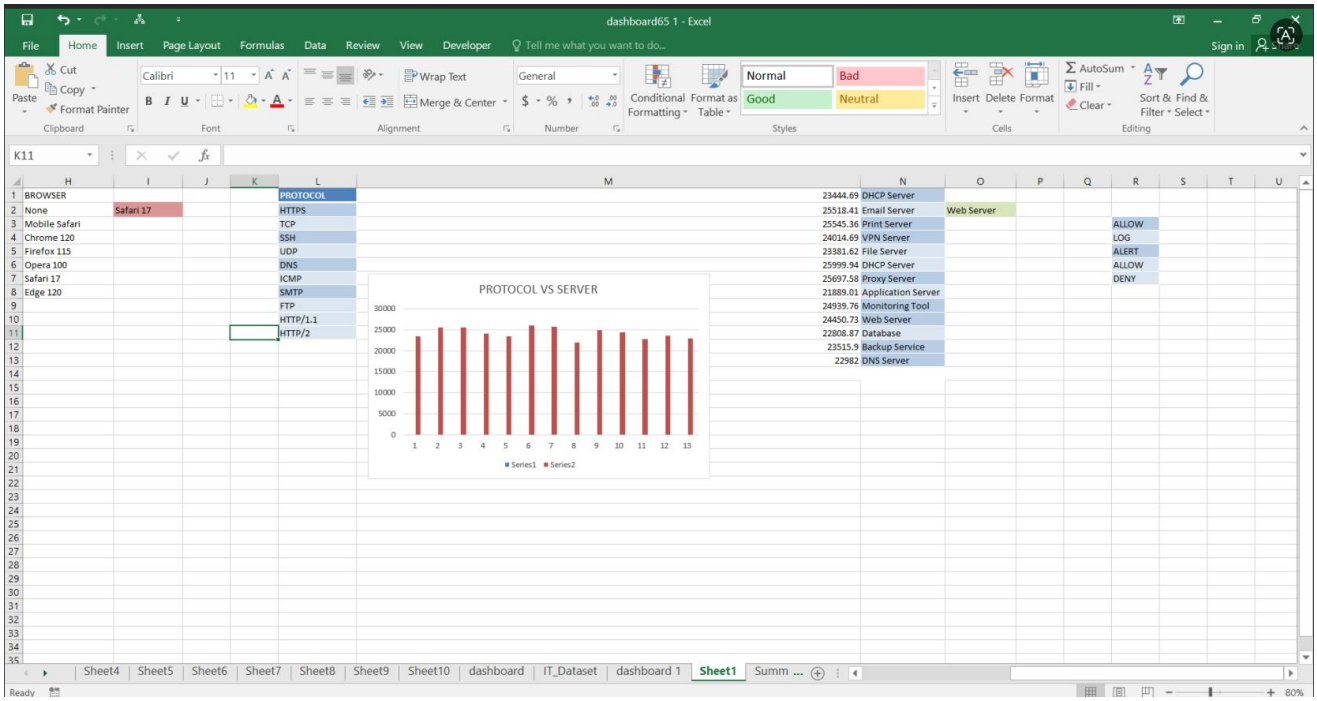
The Excel dashboard provided an overview of network performance and IT infrastructure through interactive visualizations.

The dashboard helped identify:

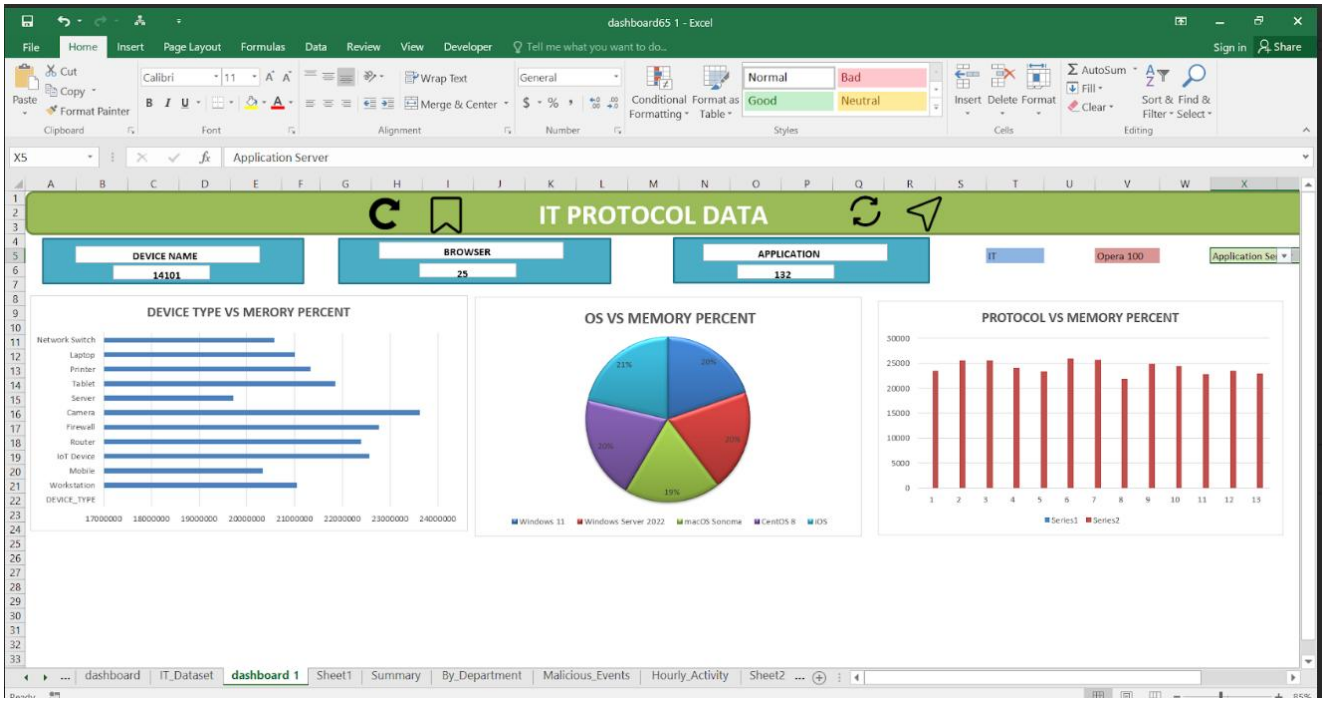
- Distribution of firewall actions.
- CPU and memory utilization.
- Packet loss percentage.
- Connection loss statistics.
- Threat score analysis.
- Department-wise performance.
- Operating system distribution.
- Protocol-wise network traffic.

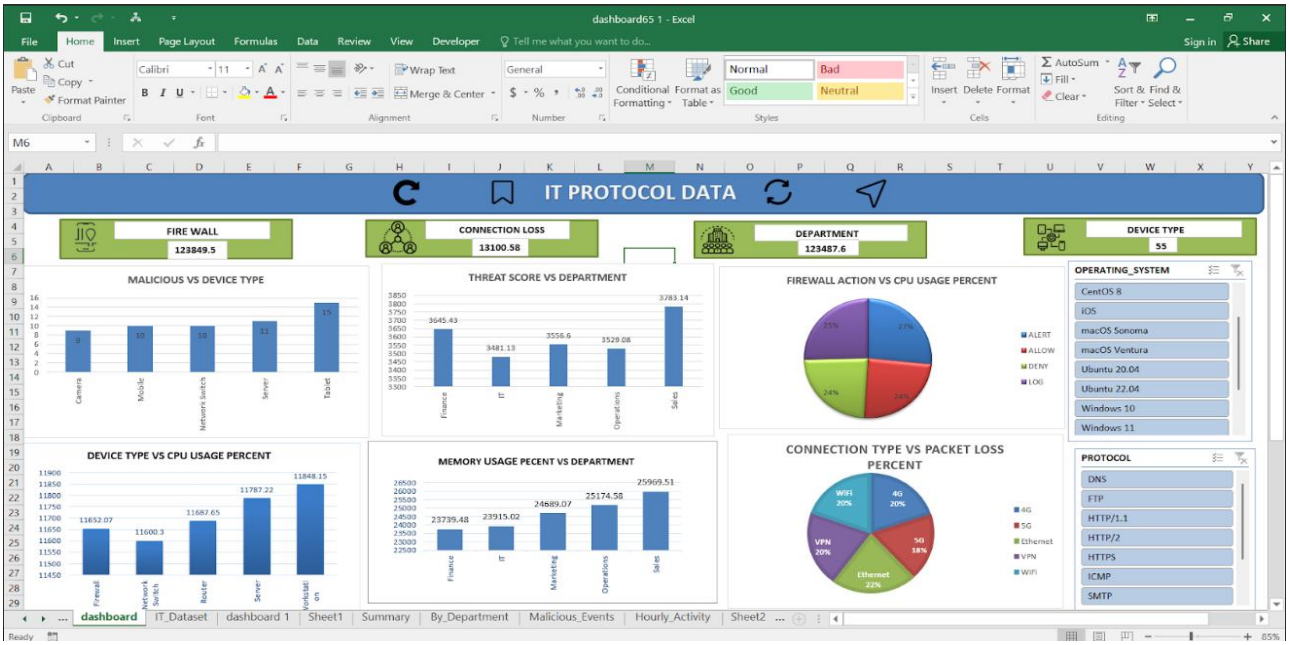
The **IT Protocol Data Dashboard** is an interactive dashboard designed to provide a quick overview of IT infrastructure and network protocol information. It presents important metrics such as the total number of devices, browsers, and applications, along with graphical representations of memory usage across different device types, operating systems, and network protocols.

The dashboard uses bar charts, pie charts, and column charts to make complex data easy to understand. It helps users identify memory consumption patterns, compare operating system usage, and monitor protocol performance. By displaying the information in a visual format, the dashboard enables faster analysis, supports informed decision-making, and helps IT administrators monitor system performance efficiently.



The dashboard uses bar charts, pie charts, and column charts to make complex data easy to understand. It helps users identify memory consumption patterns, compare operating system usage, and monitor protocol performance. By displaying the information in a visual format, the dashboard enables faster analysis, supports informed decision-making, and helps IT administrators monitor system performance efficiently.





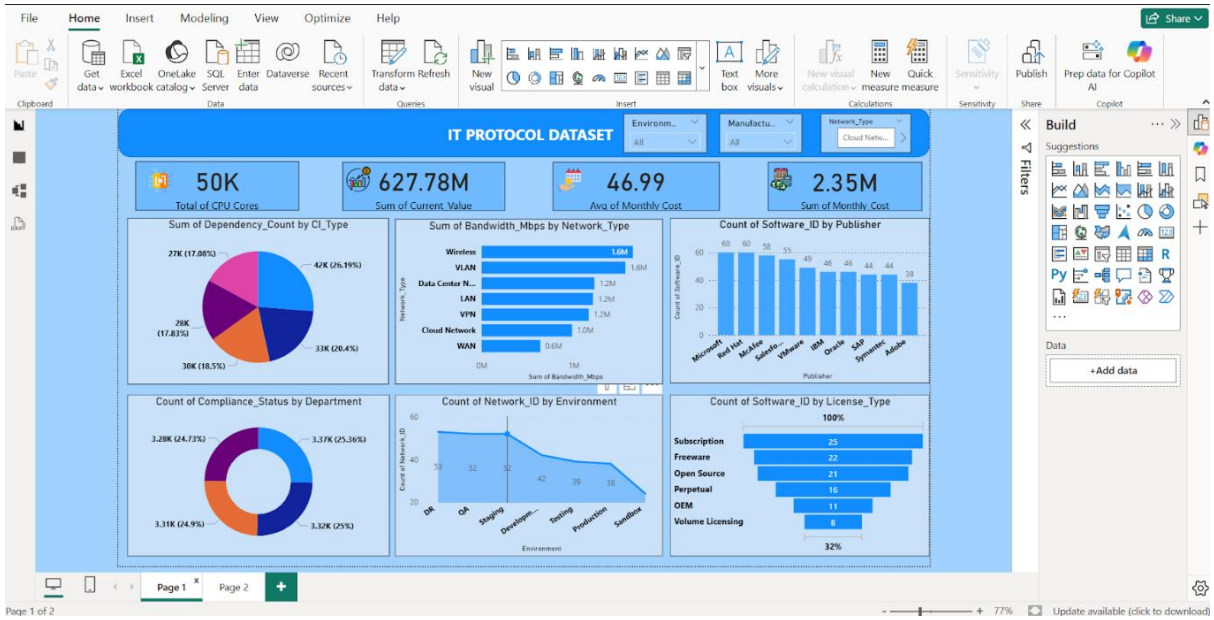
The use of Pivot Tables, Pivot Charts, and slicers allowed users to explore the data efficiently.

Power BI Dashboard Analysis

The Power BI dashboard provided advanced business intelligence features and interactive visualizations. It enabled detailed analysis of software assets, network environments, bandwidth utilization, compliance status, and operational costs.

The dashboard highlighted:

- Total CPU Cores available across systems.
- Current asset values.
- Monthly operational costs.
- Software licensing distribution.
- Compliance status of software.
- Software publisher distribution.
- Network environment categories.
- Bandwidth usage patterns.

- 

Interactive filters and slicers allowed users to analyze data from multiple perspectives and quickly identify trends and performance indicators.

Overall Results

The project successfully demonstrated the complete Data Analytics process, starting from database management using SQL, followed by data cleaning in Excel, and concluding with dashboard development in Excel and Power BI.

The developed dashboards improved data visualization, reduced manual reporting effort, and provided meaningful business insights. The internship enhanced practical knowledge in SQL, Microsoft Excel, and Power BI while strengthening analytical thinking and problem-solving skills.

CHAPTER 5
CONCLUSION

The internship in **Data Analytics** provided an excellent opportunity to gain practical knowledge and hands-on experience in data management, data analysis, and business intelligence. Throughout the internship, I learned how to transform raw data into meaningful information using industry-standard tools such as **MySQL Workbench, Microsoft Excel, and Microsoft Power BI**.

The project involved analyzing an **IT Protocol Dataset**, which contained information related to network protocols, software details, departments, operating systems, CPU usage, bandwidth utilization, software licensing, firewall actions, and other IT infrastructure data. The project followed a structured workflow, beginning with database creation and SQL query execution in MySQL Workbench, followed by data cleaning and preprocessing in Microsoft Excel, and finally developing interactive dashboards using Microsoft Excel and Microsoft Power BI.

Using **SQL**, I learned to create databases and tables, insert and update records, retrieve information through queries, and apply advanced SQL concepts such as joins, aggregate functions, grouping, sorting, and subqueries. These operations improved my understanding of relational databases and efficient data management.

In **Microsoft Excel**, I performed data cleaning, organized the dataset, created Pivot Tables, Pivot Charts, slicers, and KPI cards, and developed an interactive dashboard for analyzing key performance indicators. This helped me understand the importance of accurate data preparation and visualization.

Using **Microsoft Power BI**, I created professional dashboards by applying Power Query for data transformation, data modeling, relationships, and DAX measures. Interactive visualizations such as bar charts, pie charts, line charts, donut charts, funnel charts, KPI cards, and slicers enabled detailed analysis of the IT Protocol Dataset and provided meaningful business insights.

The dashboards developed during the internship successfully presented important information related to network performance, software distribution, compliance status, bandwidth utilization, operational costs, and system resources. These visual reports make it easier for users to monitor performance, identify trends, and support data-driven decision-making.

Overall, this internship significantly enhanced my technical knowledge and practical skills in **SQL, Microsoft Excel, and Power BI**. The experience gained during this internship has strengthened my confidence and prepared me for future career opportunities in the fields of **Data Analytics, Business Intelligence, and Data Visualization**.

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